A question that evidence can answer

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for A question that evidence can answer before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Engineering demonstration and defense

Without opening the old lesson, explain one key idea from Engineering demonstration and defense. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported A question that evidence can answer trace and record the first step you can explain without help.

Use W2: Trace the worked case for A question that **evidence** can answer. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed A question that evidence can answer explanation against the supplied facts before discussing it.

Use the worked case for A question that **evidence** can answer. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied A question that evidence can answer facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied A question that evidence can answer facts and keep your reasoning.

Proposal says prove AI is useful; available records are retrieval results for five room queries. Show the intermediate steps and give the conclusion with its **scope**.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied A question that evidence can answer facts and keep your reasoning.

A report omits this qualification: A measurable narrow question is not **evidence** of its answer. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied A question that evidence can answer facts and keep your reasoning.

You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied A question that evidence can answer facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected **outcome** before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect A question that evidence can answer to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled a question that **evidence** can answer **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a question that **evidence** can answer.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: A question that evidence can answer

> A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks. Explain the labelled step before checking the changed condition.

A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

AI ACADEMY / AI-7/1

A question that evidence can answer

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Narrow the question

Name versions A and B, the six tasks and correct-answer scoring.



2. Define the comparison

Use the same tasks and labels with and without the diagram.



3. Bound the conclusion

The study can compare those versions on those tasks, not all diagrams or learners.

CHANGE ONE THING

Replace the six tasks with six easier tasks only for B.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which proposed diagram question has a measurable **outcome**? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The study can compare those versions on those tasks, not all diagrams or learners.

1. Define the comparison

2. Bound the conclusion

3. Narrow the question

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound the conclusion The study can compare those versions on those tasks, not all diagrams or learners.

An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace the six tasks with six easier tasks only for B. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The comparison no longer isolates the diagram condition.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The comparison no longer isolates the diagram condition. Task difficulty now changes alongside the proposed intervention.

Repair a missing building block

Review Engineering demonstration and defense (ai-6/36). Goal: Explain the **mechanism** and terminology of engineering demonstration and defense. Short practice: Locate V7, room E and read-only **scope** in the demonstration.

Which **evidence** supports the answer, and why must a request to email everyone still be refused?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

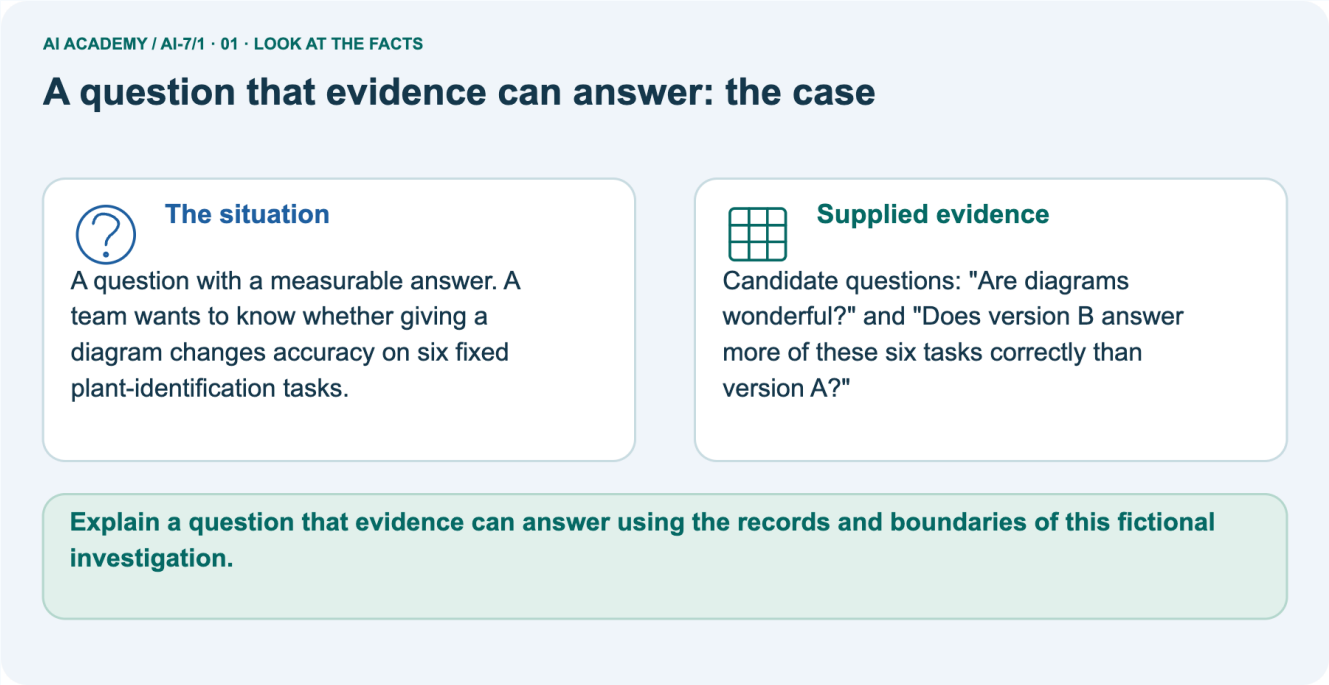
> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each A question that evidence can answer picture from its labels, then record a relevant limit.

01. A question that evidence can answer: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

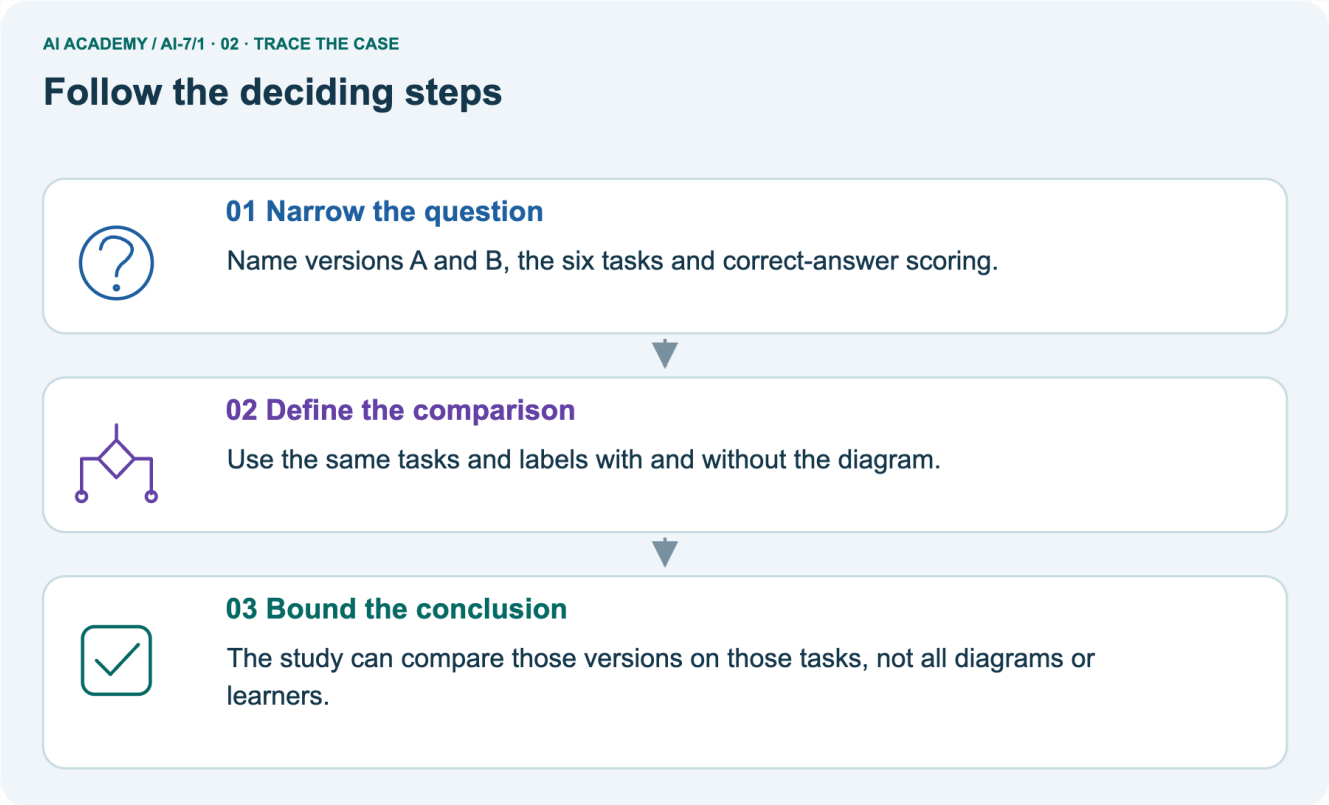
A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Supplied facts: Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



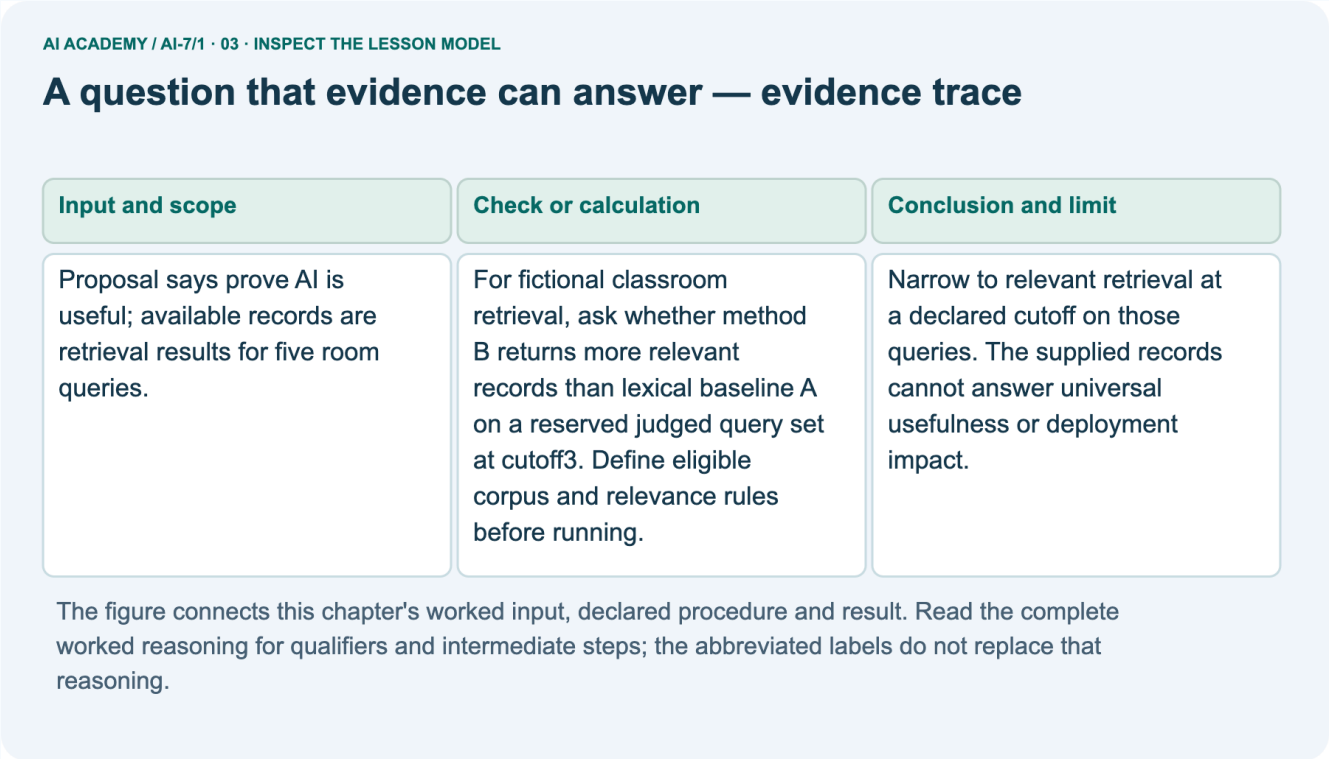
Follow the labelled picture with the matching explanation. Open the vector diagram

Narrow the question: Name versions A and B, the six tasks and correct-answer scoring. → Define the comparison: Use the same tasks and labels with and without the diagram. → Bound the conclusion: The study can compare those versions on those tasks, not all diagrams or learners.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A question that evidence can answer — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

A question that **evidence** can answer — **evidence** trace.

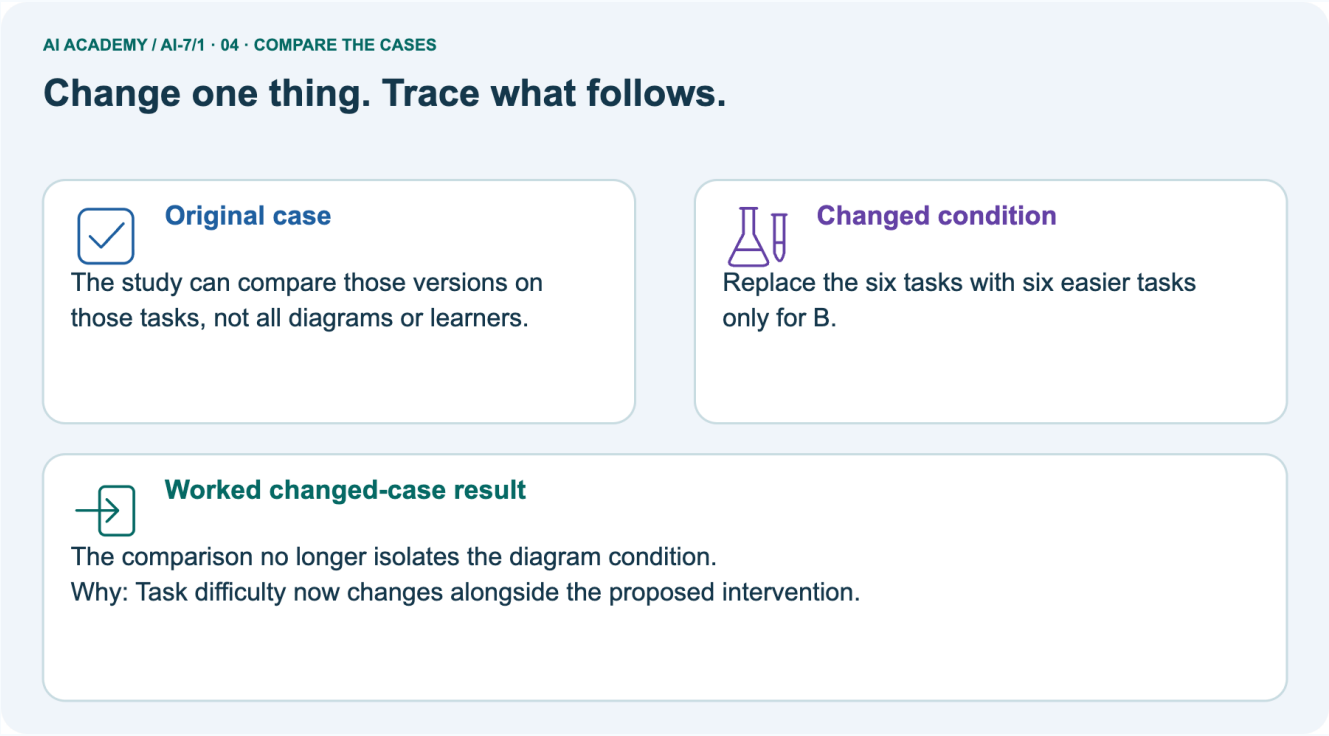
Proposal says prove AI is useful; available records are retrieval results for five room queries. | For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff3.

Define eligible corpus and relevance rules before running. | Narrow to relevant retrieval at a declared cutoff on those queries. The supplied records cannot answer universal usefulness or deployment impact.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

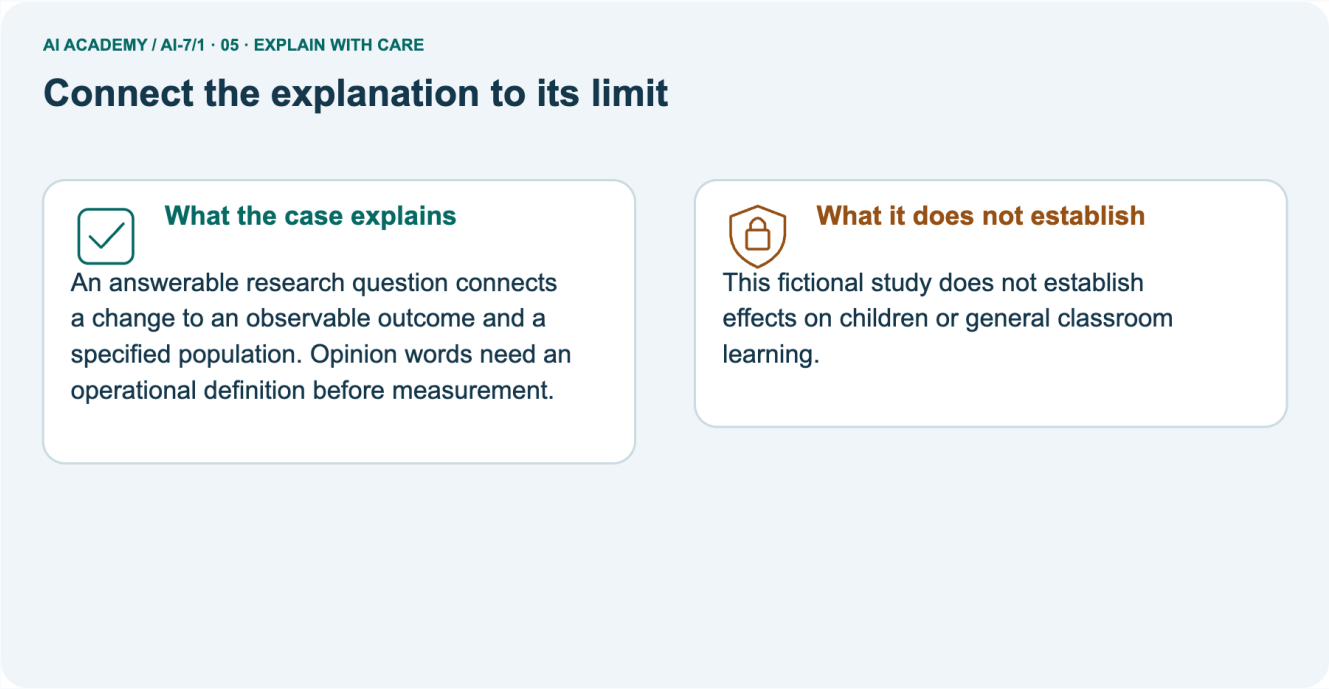
Original case: The study can compare those versions on those tasks, not all diagrams or learners.
Changed condition: Replace the six tasks with six easier tasks only for B.

Expected result: The comparison no longer isolates the diagram condition. Reason: Task difficulty now changes alongside the proposed intervention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

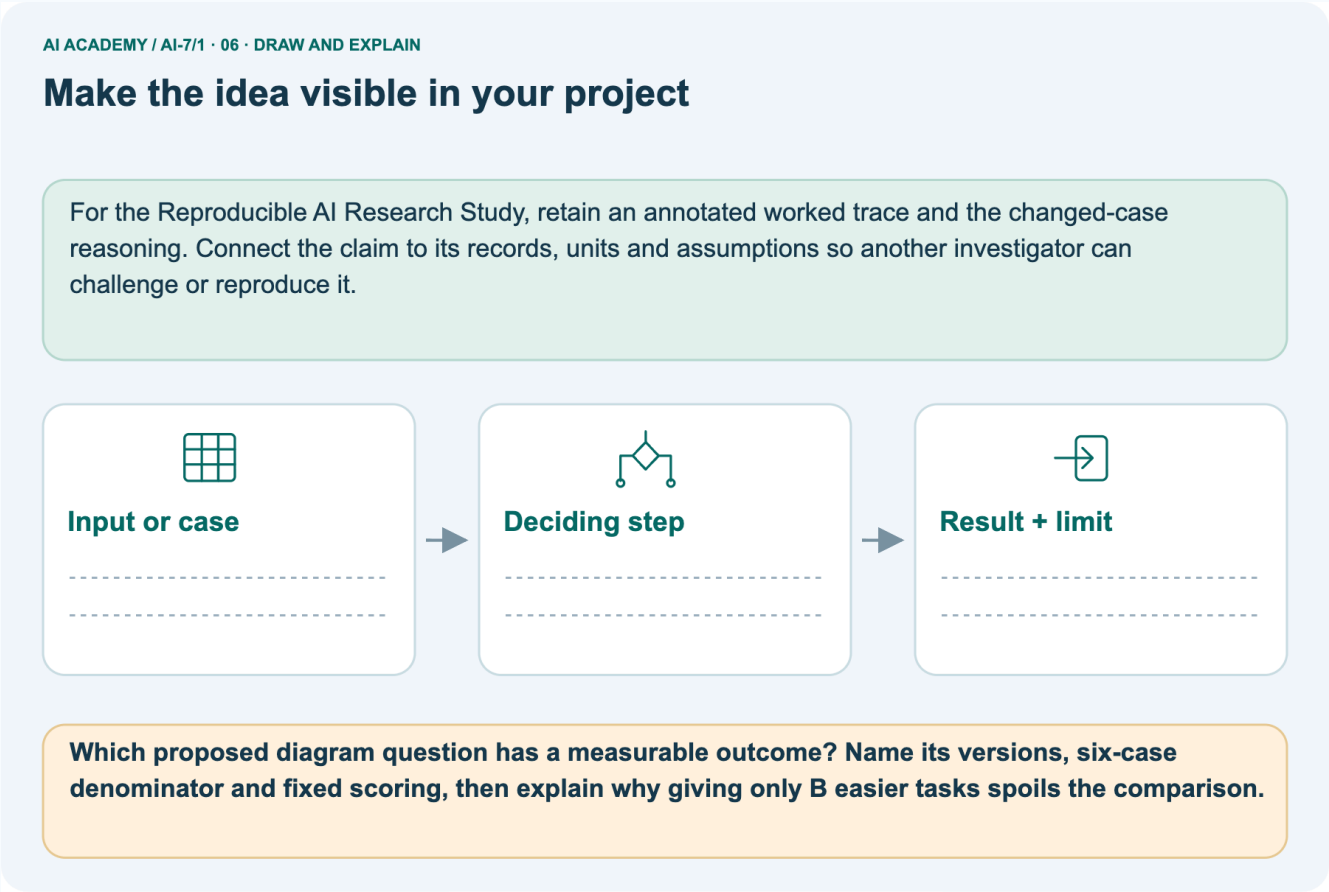
An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Limit: This fictional study does not establish effects on children or general classroom learning.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Which proposed diagram question has a measurable **outcome**?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.